

1) Atomic and molecular structure:

- a) Bohr's model of atomic structure & related numerical to determine the radius of orbit, ionization energy.
- b) Concept of wave-particle dual nature (de Broglie concept) of an electron & related numerical.
- c) Concept of Heisenberg Uncertainty Principle & related numerical.
- d) Derive the one-dimensional, time-independent Schrödinger wave equation.
- e) Derive energy expression for a particle in 1D box & related numerical.
- f) Concept of different atomic quantum numbers & their relation with each other.
- g) Concept of Radial Distribution Curves for different atomic orbitals.
- h) Concept of formation of molecular orbitals (bonding & anti-bonding molecular orbitals), Draw molecular orbital (MO)-diagram for H_2 , H_2^+ , He_2 , He_2^+ , He_2^{2+} , Be_2 , C_2 , C_2^+ , C_2^- , N_2 , N_2^+ , N_2^- , O_2 , O_2^+ , O_2^- , O_2^{2-} , F_2 , F_2^+ , F_2^- molecules & calculate bond order, magnetic moment for each molecule. Concept of Bond energy, Stability of the molecule related to bond order, concept of para & diamagnetism related to magnetic moment value.
- i) Concept of splitting of metal d-orbitals in presence of octahedral and tetrahedral field of ligand & corresponding CFSE as well as magnetic moment calculation for a coordination compounds based on nature of ligand (strong/weak field).

2) Spectroscopic techniques and applications:

- a) Difference between Emission & Absorption spectroscopy.
- b) UV-VIS:
 - i) Definition of chromophore, Auxochrome, Bathochromic shift, Hypsochromic shift, Hyperchromic shift, Hypochromic shift.
 - ii) Designation of UV absorption bands.
 - iii) UV absorption of conjugated systems.
 - iv) Effect of solvent.
- c) IR:
 - i) Conditions of IR absorption.
 - ii) Different types of bond stretching.
 - iii) Memorize stretching frequency values for the bonds present in organic functional groups.
- d) NMR:
 - i) Which nuclei exhibit the NMR phenomenon, explain with basic theory.
 - ii) Concept of Chemical Shift & factors (like shielding or deshielding effect) affecting chemical shift.
 - iii) Reasons for taking tetramethylsilane (TMS) as reference compound.
- e) Draw Jablonski diagram and explain each electronic state, each electronic transition in comparative manner.

- f) Derive the expression for the rotational energy levels of a rigid diatomic molecule, & related numericals. State and justify the selection rules for rotational spectroscopy.
- g) Obtain an expression for the energy levels of a quantum mechanical harmonic oscillator by solving the Schrödinger wave equation. State and justify the selection rules for vibrational spectroscopy.

3) Intermolecular forces and potential energy surfaces:

- a) Derivation of the Van der Waals gas equation.
- b) Derive the expressions for the critical temperature (T_c), critical pressure (P_c), and critical volume (V_c) in terms of the Van der Waals constants a , b and R .
- c) Draw a typical set of Andrews isotherms for (CO_2) (P vs. V at different T). Label the regions corresponding to gas phase, liquid phase, solid phase, gas-liquid coexistence, and the critical point. Explain how the isotherms of a real gas like CO_2 differ significantly from those predicted by the ideal gas law, particularly in the condensation region.
- d) Define the Compressibility Factor (Z). What is its value for an ideal gas under all conditions of temperature and pressure? Under what specific conditions (temperature and pressure) do all real gases approach ideal behaviour, regardless of their chemical identity? What is the value of Z under these conditions?

4) Use of free energy in chemical equilibria:

- a) intensive & Extensive Property with examples.
- b) Types of thermodynamic systems with examples.
- c) First law of thermodynamics, concept of enthalpy, concept of C_p & C_v , Determination of work done during Reversible as well as Irreversible Isothermal & Adiabatic process.
- d) Second law of thermodynamics.
- e) Criteria of spontaneity of a reaction in terms of Gibbs energy (G), Helmholtz free energy (A), entropy (S).
- f) All differential equations between the above parameters (G , A , S , P , T , V), Maxwell relations determination.
- g) Entropy calculation for ideal closed system (for Reversible as well as Irreversible Isothermal & Adiabatic process).
- h) Calculation of Gibbs energy & entropy change during mixing of gases.
- i) Concept of electrochemical series and its application to set up a galvanic cell & corresponding standard cell potential calculation. Role of Salt-Bridge.
- j) Calculation of formal cell potential using Nernst eqn.
- k) Gibbs free energy calculation using cell potential.
- l) Faraday 1st & 2nd law for electrolytic cells and related numericals.

- m) Concept of conductance, specific & equivalent or molar conductivity with units, their variation with dilution for both strong & weak electrolytes, Definition of transport number, ionic mobility.
- n) Kohlrausch dilution law & its application to calculate the equivalent or molar conductivity at infinite dilution.
- o) Definition & examples of primary, secondary, fuel cell.
- p) pH calculation of weak acid's aqueous solution.
- q) Significance of Ostwald's dilution law.
- r) Solubility product determination for sparingly soluble salts.
- s) Relation between ionic & solubility product.
- t) Concept of common-ion effect with example.
- u) Concept of buffer and its pH calculation using Henderson equation.
- v) Concept of buffer capacity.

5) Periodic properties:

Revise the periodic table chapter from Class 11 once.

6) Stereochemistry:

- a) Types of Structural & Stereo isomers with examples.
- b) Concept of chirality of a molecule.
- c) Differentiate Homomers, Enantiomers (Optical isomers) & Diastereomers with suitable examples.
- d) R/S nomenclature of chiral centres present in enantiomers.
- e) E/Z nomenclature of geometrical isomers (those alkenes which can't be identified by cis-trans isomerism).

7) Organic reactions and synthesis of a drug molecule:

- a) Concept of Inductive effect, Resonance, Hyperconjugation with examples.
- b) Structure, hybridization & Stability of carbocations, carbanions & free radicals with examples.
- c) S_N1 , S_N2 , E1, E2 - Reaction mechanisms, Factors affect these reactions' rate, Stereochemical outcomes with suitable examples.
- d) Important Reactions: Markownikoff-Anti Markownikoff addition, Aldol Condensation, Cannizaro, Hoffmann Bromamide, Diels Alder, Various Aromatic Electrophilic substitution reactions like Friedel craft, Nitration, Halogenation, Sulphonation (WITH EXAMPLES).